

Hospital Data Manager

SQL Project Report — PostgreSQL

This report contains the database setup, table structure, and 10 SQL analysis tasks performed on hospital data. The queries demonstrate aggregation, grouping, sorting, date calculations, filtering, and monthly reporting.

Skills Demonstrated

CREATE DATABASE • CREATE TABLE • DROP TABLE • SELECT • SUM • AVG • GROUP BY • ORDER BY • LIMIT •
NULLIF • DATE arithmetic • DATE_TRUNC • TO_CHAR

Database and Table Creation

```
-- Create database
CREATE DATABASE hospitaldatamanager;

-- Create table
DROP TABLE IF EXISTS hospital_data;

CREATE TABLE hospital_data (
    Hospital_name    VARCHAR(100),
    location         VARCHAR(100),
    Department       VARCHAR(100),
    Doctors_Count    INT,
    Patients_Count   INT,
    Admission_date   DATE,
    Discharge_Date   DATE,
    Medical_Expenses NUMERIC(10,2)
);

SELECT * FROM hospital_data;
```

1. Total Number of Patients

Calculates the total patient count across all hospital records.

```
-- Find the total number of patients across all hospitals.
SELECT SUM(patients_count) AS total_patients
FROM hospital_data;
```

2. Average Number of Doctors per Hospital

Calculates the average number of doctors for each hospital.

```
-- Retrieve the average count of doctors available in each hospital.
SELECT hospital_name,
       AVG(doctors_count) AS average_doctors
FROM hospital_data
GROUP BY hospital_name;
```

3. Top 3 Departments with the Highest Number of Patients

Returns the three records with the highest patient counts.

```
-- Find the top 3 hospital departments with the highest
-- number of patients.
SELECT department,
       patients_count
FROM hospital_data
ORDER BY patients_count DESC
LIMIT 3;
```

4. Hospital with the Maximum Medical Expenses

Identifies the record with the highest medical expense value.

```
-- Identify the hospital that recorded the highest medical expenses.
SELECT hospital_name,
       medical_expenses
FROM hospital_data
GROUP BY hospital_name, medical_expenses
ORDER BY medical_expenses DESC
LIMIT 1;
```

5. Daily Average Medical Expenses

Calculates medical expenses relative to the length of stay.

```
-- Calculate the average medical expenses per day for each hospital.
SELECT hospital_name,
       AVG(
           medical_expenses /
           NULLIF(discharge_date - admission_date, 0)
       ) AS avg_medical_expense_per_day
```

```
FROM hospital_data
GROUP BY hospital_name
ORDER BY avg_medical_expense_per_day DESC;
```

Note: NULLIF prevents division by zero when admission and discharge dates are the same.

6. Longest Hospital Stay

Calculates the stay duration and returns the longest stay record.

```
-- Find the record with the longest hospital stay.
SELECT hospital_name,
       patients_count,
       (discharge_date - admission_date) AS patient_longest_stay
FROM hospital_data
ORDER BY (discharge_date - admission_date) DESC
LIMIT 1;
```

Correction applied: the stay duration is sorted directly, rather than sorting by patient count.

7. Total Patients Treated Per City

Aggregates patients by location/city.

```
-- Count the total number of patients treated in each city.
SELECT location,
       SUM(patients_count) AS total_patients_per_city
FROM hospital_data
GROUP BY location
ORDER BY total_patients_per_city DESC;
```

8. Average Length of Stay Per Department

Calculates the average stay duration for each department.

```
-- Calculate the average number of days patients spend in each department.
SELECT department,
       AVG(discharge_date - admission_date) AS average_length_of_stay
FROM hospital_data
GROUP BY department
ORDER BY average_length_of_stay DESC;
```

9. Department with the Lowest Number of Patients

Finds the department with the smallest total patient count.

```
-- Find the department with the least number of patients.
SELECT department,
       SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```

10. Monthly Medical Expenses Report

Groups medical expenses by admission month.

```
-- Group the data by month and calculate total medical expenses.
SELECT DATE_TRUNC('month', admission_date) AS month,
       SUM(medical_expenses) AS total_medical_expenses
FROM hospital_data
GROUP BY DATE_TRUNC('month', admission_date)
ORDER BY month;

-- Alternative format:
SELECT TO_CHAR(admission_date, 'YYYY-MM') AS month,
       SUM(medical_expenses) AS total_medical_expenses
FROM hospital_data
GROUP BY TO_CHAR(admission_date, 'YYYY-MM')
ORDER BY month;
```

Two PostgreSQL approaches: DATE_TRUNC keeps a date/timestamp value, while TO_CHAR formats the month as text such as YYYY-MM.

Project Summary

The project uses PostgreSQL to analyze hospital operations and financial data. It covers patient volume, doctor availability, hospital expenses, length of stay, department-level analysis, city-level patient totals, and monthly expense reporting. These queries provide practical examples of SQL techniques commonly used in data analyst projects and interviews.